



Department of Computer Science and Engineering (Data Science)

B.Tech. Sem: III Subject: Statistics for Data Science

Experiment 6

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Date:	Experiment Title: Test of hypothesis for large samples
Aim	To study hypothesis testing for proportion and mean for large samples.
Software	Google Colab
	Solve the problems manually as well as using python Question 1: Out of 500 students, 410 students pass the exam. Test the hypothesis that more than 80% of students pass the exam. Level of significance = 5% <pre> In []: #Question 1 from scipy.stats import norm p_p = 0.8 sample_success = 410 sample_size = 500 from statsmodels.stats.proportion import proportions_ztest stat, p_value = proportions_ztest(count=sample_success, nobs = sample_size, value = print('z_stat = ', stat, 'p_value = ', p_value) z_cri = norm.ppf(0.95) print('z_cri = ', z_cri) z_stat = 1.1640504929492914 p_value = 0.12220177493249235 z_cri = 1.6448536269514722 In []: s_p = sample_success/sample_size SE = np.sqrt(p_p*(1-p_p)/sample_size) z = (s_p-p_p)/SE print('z_stat = ', z) z_cri = norm.ppf(0.95) p = norm.sf(z) </pre> Since, $z < z_{\text{critical}}$, null hypothesis is not rejected & alternate hypothesis is rejected. Hence, 80% of the students have passed the exam. Question 2: A random sample of 35 individuals selected from a certain population showed that 6 of them are left-handed. Find 95% and 99% confidence limits for the proportions of left-handed person in the population.

```
In [ ]: #Question 2
import numpy as np
s_p = 6/35
n = 35
SE = np.sqrt(s_p*(1-s_p)/n)
z = norm.ppf(0.975)
L = s_p - z*SE
U = s_p + z*SE
CI = (L,U)
print('95% Confidence Interval for population means = ',CI)
z = norm.ppf(0.995)
L = s_p - z*SE
U = s_p + z*SE
CI = (L,U)
print('99% Confidence Interval for population means = ',CI)
```

```
95% Confidence Interval for population means = (0.04656934249155212, 0.29628780036559077)
99% Confidence Interval for population means = (0.007335729288939441, 0.3355214135682034)
```

Question 3: The manufacturer of patent medicine claimed that it was 90% effective in relieving allergy for a period of 8 hours. In a sample of 200 people who had the allergy, the medicine provided relief for 160 people. Determine whether manufacturer's claim is legitimate.

```
In [ ]: #Question 3
#Ho : P = 0.9 , Ha != 0.9
sample_size = 200
sample_success = 160
p_p = 0.9
stat,p_value = proportions_ztest(count=sample_success,nobs = sample_size,value =
z_cri = norm.ppf(0.95)
print('z_stat = ',stat,'p_value = ',p_value)
print('z_cri = ',z_cri)
```

```
z_stat = -3.535533905932737 p_value = 0.00040695201744495973
z_cri = 1.6448536269514722
```

Question 4: From one class of 500 students, 410 students passed. From other class of 400 students, 379 students passed. Test the hypothesis that both samples have same proportions. Level of significance = 0.025

```
In [ ]: alpha = 0.025
success = np.array([410,379])
sample = np.array([500,400])
stat,p_value = proportions_ztest(count=success,nobs = sample,alternative = 'two-s
if p_value > alpha:
    print("Test is not significant.Fail to reject null hypothesis")
else:
    print("Test is significant.Reject null hypothesis")
print('z_stat = ',stat,'p_value = ',p_value)
```

```
Test is significant.Reject null hypothesis
z_stat = -5.7802476568050825 p_value = 7.459074060078635e-09
```

Question 5: Two groups, A and B, each consists of 100 people who have a disease. A serum is given to group A but not group B; otherwise both the groups are treated identically. It's

found that in group A and group B, 75 and 65 people respectively, recover from the disease. Test the hypothesis that serum helps to cure the disease using a level of significance (a) 0.01 (b) 0.1.

```
In [ ]: alpha = 0.01
success = np.array([75,65])
sample = np.array([100,100])
stat,p_value = proportions_ztest(count=success,nobs = sample,alternative = 'large')
if p_value > alpha:
    print("Test is not significant.Fail to reject null hypothesis")
else:
    print("Test is significant.Reject null hypothesis")
print('z_stat = ',stat,'p_value = ',p_value)
```

Test is not significant.Fail to reject null hypothesis
z_stat = 1.5430334996209185 p_value = 0.06141132405069633

```
In [ ]: alpha = 0.1
success = np.array([75,65])
sample = np.array([100,100])
stat,p_value = proportions_ztest(count=success,nobs = sample,alternative = 'large')
if p_value > alpha:
    print("Test is not significant.Fail to reject null hypothesis")
else:
    print("Test is significant.Reject null hypothesis")
print('z_stat = ',stat,'p_value = ',p_value)
```

Test is significant.Reject null hypothesis
z_stat = 1.5430334996209185 p_value = 0.06141132405069633

Question 6: Check the hypothesis that blood pressure before treatment has a mean value of 156 for the given dataset.

```
In [ ]: import pandas as pd
df = pd.read_csv("/content/blood_pressure.csv")
df.head()
```

Out[16]:

	patient	sex	agegrp	bp_before	bp_after
0	1	Male	30-45	143	153
1	2	Male	30-45	163	170
2	3	Male	30-45	153	168
3	4	Male	30-45	153	142
4	5	Male	30-45	146	141

```
In [ ]: df.describe()
```

```
Out[17]:
```

	patient	bp_before	bp_after
count	120.000000	120.000000	120.000000
mean	60.500000	156.450000	151.358333
std	34.785054	11.389845	14.177622
min	1.000000	138.000000	125.000000
25%	30.750000	147.000000	140.750000
50%	60.500000	154.500000	149.500000
75%	90.250000	164.000000	161.000000
max	120.000000	185.000000	185.000000

```
In [ ]: alpha = 0.05
from statsmodels.stats import weightstats as stests
z_stats,p_val = stests.ztest(df['bp_before'],x2 = None,value = 156,altern
print('Z_stat:',z_stats,'p value=',p_val)
print('z_critical=',norm.ppf(0.975))
if p_val > alpha:
    print("Test is not significant.Fail to reject null hypothesis")
else:
    print("Test is significant.Reject null hypothesis")
```

```
Z_stat: 0.432798073526164 p value= 0.6651614730255063
z_critical= 1.959963984540054
Test is not significant.Fail to reject null hypothesis
```

Question 7: Can we say that the systolic blood pressure measured on arm has mean value of 125 mmHg from the given dataset?

```
In [ ]: df = pd.read_csv('/content/systolic blood pressure.csv')
df.head()
```

```
Out[24]:
```

	armsys	fingsys
0	140	154
1	110	112
2	138	156
3	124	152
4	142	142

In []: df.describe()

Out[26]:

	armsys	fingsys
count	200.000000	200.000000
mean	128.520000	132.815000
std	23.287575	25.648195
min	79.000000	60.000000
25%	111.500000	118.000000
50%	125.000000	130.000000
75%	140.000000	146.500000
max	220.000000	228.000000

```
In [ ]: alpha = 0.05
z_stats,p_val = stests.ztest(df['armsys'],x2 = None,value = 125,alternative = 'tv
print('Z_stat:',z_stats,'p value=',p_val)
print('z_critical=',norm.ppf(0.975))
if p_val > alpha:
    print("Test is not significant.Fail to reject null hypothesis")
else:
    print("Test is significant.Reject null hypothesis")
```

Z_stat: 2.1376342148925 p value= 0.032546442541153964
 z_critical= 1.959963984540054
 Test is significant.Reject null hypothesis

Question 8: A sample of 50 pieces of certain type of string was tested. The mean breaking strength turned out to be 14.5 pounds. Test whether the sample is from a batch of string having mean breaking strength of 15.6 pounds and standard deviation of 2.2 pounds.

```
In [ ]: x_bar = 14.5
mean = 15.6
sigma = 2.2
n = 50
SE = sigma/np.sqrt(n)
z =(x_bar-mean)/SE
print('Z_stat=',z)
print('Z_critical=',norm.ppf(0.025))
p_value = 2*norm.cdf(z)
print('p_value=',p_val)
if p_val > alpha:
    print("Test is not significant.Fail to reject null hypothesis")
else:
    print("Test is significant.Reject null hypothesis")
```

Z_stat= -3.5355339059327364
 Z_critical= -1.9599639845400545
 p_value= 0.032546442541153964
 Test is significant.Reject null hypothesis

Question 9: Check the hypothesis that mean value of systolic blood pressure measured on arm and finger are the same.



In []: df.describe()

Out[29]:

	armsys	fingsys
count	200.000000	200.000000
mean	128.520000	132.815000
std	23.287575	25.648195
min	79.000000	60.000000
25%	111.500000	118.000000
50%	125.000000	130.000000
75%	140.000000	146.500000
max	220.000000	228.000000

```
In [ ]: alpha = 0.05
z_stats,p_val = stests.ztest(df['armsys'],x2 = df['fingsys'],value = 0,alternativ
print('Z_stat:',z_stats,'p value=',p_val)
print('z_critical=',norm.ppf(0.025))
if p_val > alpha:
    print("Test is not significant.Fail to reject null hypothesis")
else:
    print("Test is significant.Reject null hypothesis")
```

Z_stat: -1.7533233440871472 p value= 0.07954652069053099

z_critical= -1.9599639845400545

Test is not significant.Fail to reject null hypothesis

Question 10: A common perception about COVID-19 is that warm climate is more resistant to the corona outbreak. Verify this using hypothesis testing.

```
In [ ]: corona = pd.read_csv('/content/Corona_Updated.csv')
corona.head()
```

Out[33]:

	Province/State	Country/Region	Last Update	Confirmed	Deaths	Recovered	Latitude
0	Hubei	Mainland China	2020-03-10T15:13:05	67760	3024	47743	30.9756
1	NaN	Italy	2020-03-10T17:53:02	10149	631	724	43.0000
2	NaN	Iran (Islamic Republic of)	2020-03-10T19:13:20	8042	291	2731	32.0000
3	NaN	Republic of Korea	2020-03-10T19:13:20	7513	54	247	36.0000
4	NaN	France	2020-03-10T18:53:02	1784	33	12	47.0000

```
In [ ]: corona.describe(
)
```

Out[34]:

	Confirmed	Deaths	Recovered	Latitude	Longitude	Temperature	Humidity
count	206.000000	206.000000	206.000000	206.000000	206.000000	206.000000	206.000000
mean	575.640777	20.689320	312.640777	31.184989	11.752030	12.161165	67.728155
std	4822.697784	215.794845	3332.764713	21.305149	84.576291	10.229763	21.780588
min	0.000000	0.000000	0.000000	-41.454500	-157.498300	-21.900000	6.000000
25%	3.000000	0.000000	0.000000	25.069200	-74.841325	6.100000	55.000000
50%	12.000000	0.000000	0.000000	36.030550	15.234250	11.750000	73.000000
75%	75.750000	1.000000	4.000000	43.870250	101.363375	20.375000	84.000000
max	67760.000000	3024.000000	47743.000000	64.963100	174.886000	33.100000	98.000000

```
In [ ]: corona['Temp_cat'] = corona['Temperature'].apply(lambda x:0 if x<24 else 1)
corona_t = corona[['Confirmed', 'Temp_cat']]
corona_t.head()
d1 = corona_t[(corona_t['Temp_cat']==1)]['Confirmed']
d2 = corona_t[(corona_t['Temp_cat']==0)]['Confirmed']
m1 = d1.mean()
m2 = d2.mean()
s1 = d1.std()
s2 = d2.std()
n1 = d1.shape[0]
n2 = d2.shape[0]
print('Mean confirmed cases for warm climate : ',m1)
print("Mean confirmed cases for not warm climate: ",m2)
```

Mean confirmed cases for warm climate : 26.548387096774192
 Mean confirmed cases for not warm climate: 672.9085714285715

```
In [ ]: #Ho: mu1 = mu2 , Ha : mu1<mu2
SE = np.sqrt(s1**2/n1+s2**2/n2)
z = (m1-m2)/SE
p_val = norm.cdf(z)
print(z,p_val)
if(p_val<0.05):
    print("Test is significant.Reject null hypothesis")
else:
    print("Test is not significant.Fail to Reject null hypothesis")

-1.6349753148440753 0.05102711125709931
Test is significant.Reject null hypothesis
```

Question 11: The mean consumption of food grains among 400 samples middle class consumers is 380 grams per day per person with a standard deviation of 120 grams. A similar sample survey of 600 working class consumers gave a mean of 410 grams with standard deviation of 80 grams. Are we justified in saying that the difference between the averages of the two classes is 40? Use 5% level of significance.



```
In [ ]: #Ho: |m1-m2|=40, Ha: |(m1-m2)| != 40
n1 = 400
m1 = 380
s1 = 120
n2 = 600
m2 = 410
m = np.abs(m1-m2)
s2 = 80
alpha = 0.05
SE = np.sqrt(s1**2/n1+s2**2/n2)
z = (m-40)/SE
p_val = 2*norm.cdf(z)
print('Z_stat:',z_stats,'p value=',p_val)
if(p_val<0.05):
    print("Test is significant.Reject null hypothesis")
else:
    print("Test is not significant.Fail to Reject null hypothesis")
```

Question 12: Two populations have same mean but the standard deviation of one is twice that of the other. Show that in samples, each of size 500, drawn under simple random conditions, the difference of the means will not exceed 0.3σ , where σ is the smaller standard deviation.

```
In [ ]: """
mu1 = mu2, sigma2 = 2*sigma1,n1=n2=500
z = (X1-X2)/sqrt((sigma1^2/n1)+(sigma2^2/n2))
z = (X1-X2)/sqrt((sigma^2/500)+(4sigma^2)/500)
z = (X1-X2)/(sigma/10)
(X1-X2)<=0.30
(X1-X2)/(sigma/10)<=0.3sigma/(sigma/10)
z<=3
Prob (z=3)
"""
norm.cdf(3)
```

Out[48]: 0.9986501019683699

Conclusion

Test of hypothesis of large samples in terms of proportion & mean has been conducted in this experiment to justify whether the null hypothesis is rejected or not rejected.

Signature of Faculty